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Approximation of Magnetic Characteristics

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Contents

1 Introduction

For many applications, it is sufficient to use a reasonably accurate approximation of the magnetic characteristic curve of the ferromagnetic material without taking hysteresis into account, and to calculate the hysteresis losses subsequently along with the eddy current losses. The hysteresis loop of common soft magnetic materials, such as electrical steel sheets for electric machines and transformers, is relatively narrow, which justifies this approach. When using the approximation, care must be taken to ensure a continuous curve and a consistent slope so as not to distort the numerical simulations based on it.

All of the approximation formulas described here have been modified from those found in the literature, taking physical quantities and units into account.

2 Acknowledgement

I would like to express my gratitude to voestalpine AG https://www.voestalpine.com/isovac/ for granting me permission to use the measurement data for the offered grades of non-grain-oriented electrical steel to validate the approximations: https://www.voestalpine.com/isovac/en/Downloads/Datasheets

3 Definition of Variables

Magnetic field strength	H	$\frac{A}{m}$
Magnetic flux density	$B = \mu_0 \cdot \mu_r \cdot H$	$T = \frac{V \cdot s}{m^2}$
Magnetic field constant	$\mu_0 = 1{,}25663706127 \cdot 10^{-6} \approx 4\pi \cdot 10^{-7}$	$\frac{V \cdot s}{A \cdot m}$
Relative permeability	μ_r	1
Initial relative permeability	$\mu_{ri} = \mu_r(H \to 0)$	1
Differential rel. Permeability	$\mu_{rd} = \frac{1}{\mu_0} \cdot \frac{dB}{dH}$	1
Magnetic polarization	$J + \mu_0 \cdot H = B$ $J = \mu_0 \cdot M = \mu_0 \cdot \chi \cdot H$	$\frac{V \cdot s}{m^2}$
Magnetization	$M = \chi \cdot H$	$\frac{A}{m}$
Magnetic susceptibility	$\chi = \mu_r - 1$	1
Differential suszeptibility	$\chi_d = \frac{1}{\mu_0} \cdot \frac{dJ}{dH}$	1

The induced voltage can be calculated based on the assumption that an area A is traversed by the magnetic flux and the number of turns w of the coil, as follows:

$$\psi = w \cdot A \cdot B(H) = w \cdot A \cdot \mu_0 \cdot \mu_r(H) \cdot H \tag{1}$$

$$v = -\frac{d\psi}{dt} = -\frac{d\psi}{dH} \cdot \frac{dH}{dt} = -w \cdot A \cdot \frac{dB}{dH} \cdot \frac{dH}{dt} = -w \cdot A \cdot \mu_0 \cdot \mu_{rd} \cdot \frac{dH}{dt} \tag{2}$$

4 Requirements for the Approximation

The approximation formula used should be intrinsically safe, i.e., it should not yield non-physical results even under extreme extrapolation. It seems appropriate to focus on the approximation of the magnetic polarization and to exclude the linear term $\mu_0 \cdot H$ from the flux density, since the polarization exhibits a horizontal asymptote at high field strengths.

$$\lim_{H\to\infty} \chi(H) = \lim_{H\to\infty} \chi_d(H) = 0 \;\Leftrightarrow\; \lim_{H\to\infty} J(H) = J_{Sat} \tag{3}$$

$$\lim_{H\to\infty} \chi(H) = \lim_{H\to\infty} \chi_d(H) = \mu_{ri} - 1 \tag{4}$$

The literature describes many approaches to approximating magnetization curves. Some of the formulas are simple in structure and the parameters are easy to determine, but the accuracy of the approximation leaves much to be desired. Depending on the application, it is necessary to find an approximation that offers an acceptable trade-off between effort and accuracy.

The following regions can be distinguished along the magnetization characteristic (see Figure 1):

1. Starting from the origin, reversible wall displacements of the elementary magnets occur.
2. As the field strength increases, irreversible wall displacements of the elementary magnets occur.
3. Consequently, the high field strength forces the elementary magnets to rotate.
4. Once all rotational processes are completed, the flux density increases at the same rate as it does in a vacuum. This also means that the polarization approaches a horizontal asymptote J_{sat}—the saturation polarization.

The characteristics $J(H)$ and $B(H)$ are point-symmetric about the origin, while $\mu_r(H)$ and $\mu_{rd}(H)$ are symmetric about the y-axis. In this work only the right half-plane is considered. When applying the results, the respective formula should be extended to the left half-plane, if necessary, using $sign(H)$ und $|H|$.

Starting from the initial permeability μ_{ri} at the origin, both the relative permeability μ_r and the differential relative permeability μ_{rd} initially increase to a maximum, as described by Lord Rayleigh in [1] and Sixtus in [2].

Subsequently, both approach the value $\mu_r = 1$ asymptotically. This means that the magnetization characteristic curves $B(H)$ and $J(H)$ have an inflection point. The maximum susceptibility occurs at the point of greatest slope of the line from the origin to the point $J(H)$; at this point, the slope χ and the gradient χ_d coincide. Up to this point, $\chi < \chi_d$; from this point onward, $\chi > \chi_d$.

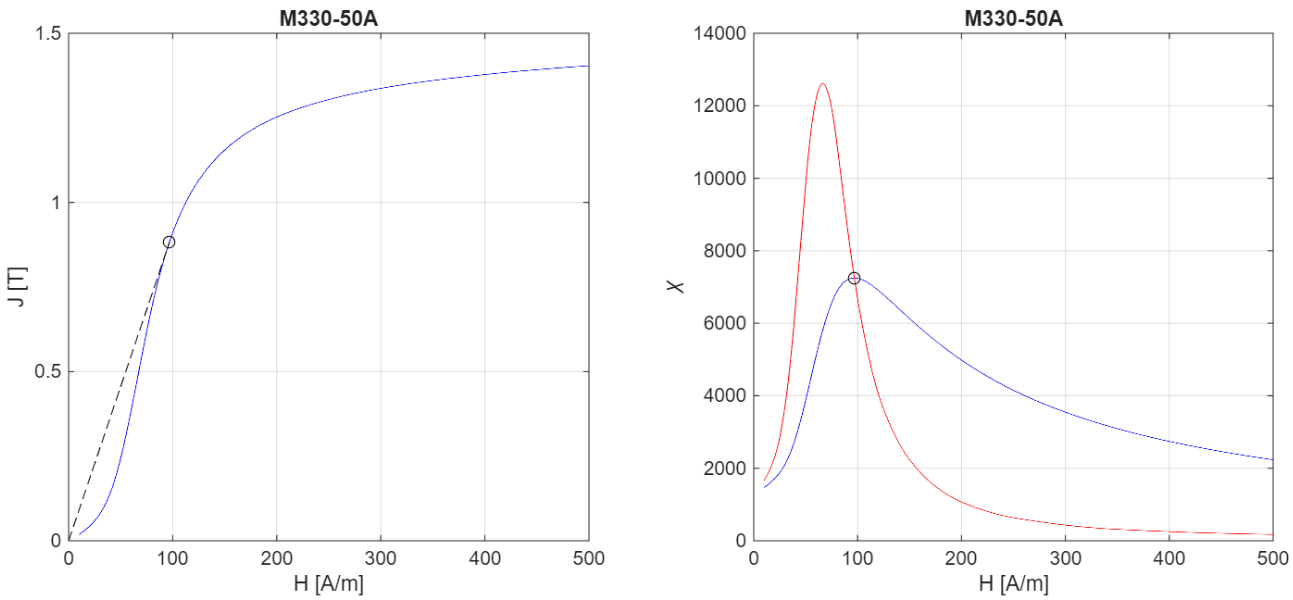

Figure 1 Magnetic characteristic and curve characteristics of relative and differential susceptibility

Determination of the maximum relative permeability:

Based on the fundamental relationship

$$B = \mu_0 \cdot \mu_r \cdot H \tag{5}$$

is obtained by differentiating and dividing by μ_0:

$$\frac{dB}{\mu_0 \cdot dH} = \mu_{rd} = \frac{d\mu_r}{dH} \cdot H + \mu_r \cdot H \tag{6}$$

$$\frac{d\mu_r}{dH} \cdot H = \mu_{rd} - \mu_r \tag{7}$$

At the maximum relative permeability, the following holds:

$$\frac{d\mu_r}{dH} = 0 \leftrightarrow \mu_{rd} = \mu_r \tag{8}$$

It may be that the details of this behavior are not critical for some applications, but they are significant for other problems. Therefore, a high-quality approximation of this region near the origin should also be sought.

The importance of a continuous approximation of the differential permeability for the subsequent calculation of the induced voltage has to be emphasized.

5 Simple Algebraic and Transcendental Functions

5.1 Approximation formulas of the form *B=f(H)*

Approximation using simple algebraic and transcendental functions has been discussed in detail in the literature, including Moser in [3] and Fischer and Moser in [4]. In such cases, compromises often have to be made regarding the accuracy of the approximation in certain regions of the characteristic curve, e.g., in the region of low magnetic field strengths.

The following formula has proven to be easy to use:

$$J(H) = \mu_0 \cdot (\mu_{ri} - 1) \cdot H_p \cdot atan\left(\frac{H}{H_p}\right) \tag{9}$$

The parameter H_p can be determined from a given point on the characteristic curve $J_N(H_N)$:

$$J_N = \mu_0 \cdot (\mu_{ri} - 1) \cdot H_p \cdot atan\left(\frac{H_N}{H_p}\right) \rightarrow H_p = H_N \cdot cot\left(\frac{J_N}{\mu_0 \cdot (\mu_{ri} - 1) \cdot H_p}\right) \tag{10}$$

$$\mu_r(H) = \frac{B(H)}{\mu_0 \cdot H} = 1 + (\mu_{ri} - 1) \cdot \frac{H_p}{H} \cdot atan\left(\frac{H}{H_p}\right) \tag{11}$$

$$\mu_{dr} = \frac{1}{\mu_0} \cdot \frac{dB}{dH} = 1 + \frac{\mu_{ri} - 1}{1 + \left(\frac{H}{H_p}\right)^2} \tag{12}$$

$$\mu_r(H \rightarrow 0) = \mu_{dr}(H \rightarrow 0) = \mu_{ri} \tag{13}$$

$$\mu_r(H \rightarrow \infty) = \mu_{dr}(H \rightarrow \infty) = 1 \tag{14}$$

This approach can easily be applied to chokes with magnetic saturation:

$$\psi(i) = L_\infty \cdot i + (L_0 - L_\infty) \cdot I_p \cdot atan\left(\frac{i}{I_p}\right) \tag{15}$$

where L_∞ denotes the inductance for $i \rightarrow \infty$ and L_0 denotes the inductance for $i \rightarrow 0$. However, the behavior at low magnetic field strengths (or currents) is not accurately represented. An advantage is that only a few data points need to be known: μ_{ri} and $J_N(H_N)$.

5.2 Approximation formulas of the form *H=f(B)*

The representation of magnetic field strength as a function of flux density is also discussed in the literature. Brauer provides a more detailed formula in [5]:

$$\mu_0 \cdot H(B) = k_1 \cdot B \cdot e^{\left(\frac{B}{B_p}\right)^2} + k_3 \cdot B \tag{16}$$

In [6], Hülsmann describes extensions of this formula for the regions of low magnetic field strengths (Rayleigh region) and high magnetic field strengths (saturation approximation).

In [7], Roschke presents a formula for relative permeability:

$$\mu_r(B) = \frac{\mu_r + c_a \cdot \frac{B}{B_{\mu Max}}}{1 + c_b \cdot \frac{B}{B_{\mu Max}} + \left(\frac{B}{B_{\mu Max}}\right)^n} \tag{17}$$

A modification of this formula was implemented in the FluxTubes Modelica library as described in [8]:

$$\mu_r(B) = 1 + \frac{\mu_{ri} - 1 + c_a \cdot \frac{B}{B_{\mu Max}}}{1 + c_b \cdot \frac{B}{B_{\mu Max}} + \left(\frac{B}{B_{\mu Max}}\right)^n} \tag{18}$$

$B_{\mu Max}$ denotes the flux density at the maximum (relative) permeability.

The difficulties in determining the parameters, the exact behavior in the region of low field strengths, and the extrapolation to high field strengths (approaching saturation) prove to be disadvantages.

6 Approximations Based on Probability Distributions

In [9], Altenbernd argues that, in accordance with the central limit theorem of statistics, the Weiss districts behave as if they were a superposition of a large number of independent factors and thus are normally distributed.

The probability density function of the normal distribution is given by:

$$pdf(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-\frac{1}{2} \cdot \left(\frac{x-\mu}{\sigma}\right)^2} \tag{19}$$

where μ represents the expected value and σ the standard deviation. Figuratively speaking, μ determines the location of the peak and σ the width of the bell curve. The density function is the derivative of the cumulative distribution function:

$$cdf(x) = \int_{-\infty}^{x} f(t) \cdot dt = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot \int_{-\infty}^{x} e^{-\frac{1}{2} \cdot \left(\frac{t-\mu}{\sigma}\right)^2} \cdot dt \tag{20}$$

If we substitute $u = \frac{x-\mu}{\sigma}$, we obtain the standard normal distribution (for $\mu = 0$ and $\sigma = 1$):

$$\varphi(u) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{u^2}{2}} \tag{21}$$

$$\phi(u) = \int_{-\infty}^{u} \varphi(t) \cdot dt = \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{u} e^{-\frac{t^2}{2}} \cdot dt \tag{22}$$

The following relationship with the error function is valid:

$$\phi(u) = \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{u} e^{-\frac{t^2}{2}} \cdot dt = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \cdot \int_{0}^{u} e^{-\frac{t^2}{2}} \cdot dt = \frac{1}{2} \cdot \left(1 + erf\left(\frac{u}{\sqrt{2}}\right)\right) \tag{23}$$

Differential susceptibility is approximated by a sum of shifted and scaled density functions:

$$\chi_d = \sum_{k=1}^{N} \chi_{dk} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\left(\frac{1}{\sqrt{2}} \cdot \frac{H - H_{\mu k}}{H_{\sigma k}}\right)^2} \tag{24}$$

Polarization can be obtained by integration:

$$J = \mu_0 \cdot \sum_{k=1}^{N} \chi_{dk} \cdot H_{\sigma k} \cdot \frac{1}{2} \cdot \left[erf\left(\frac{1}{\sqrt{2}} \cdot \frac{H - H_{\mu k}}{H_{\sigma k}}\right) - erf\left(\frac{1}{\sqrt{2}} \cdot \frac{0 - H_{\mu k}}{H_{\sigma k}}\right)\right] \tag{25}$$

The second erf term is an integration constant and ensures that $J(H = 0) = 0$.

$$\lim_{H \to 0} \chi_d = \sum_{k=1}^{N} \chi_{dk} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\left(\frac{1}{\sqrt{2}} \cdot \frac{-H_{\mu k}}{H_{\sigma k}}\right)^2} = \mu_{ri} - 1 \tag{26}$$

$$\lim_{H \to \infty} J = \mu_0 \cdot sign(H) \cdot \sum_{k=1}^{N} \chi_{dk} \cdot H_{\sigma k} \cdot \frac{1}{2} \cdot \left[1 - erf\left(\frac{1}{\sqrt{2}} \cdot \frac{-H_{\mu k}}{H_{\sigma k}}\right)\right] = J_{sat} \tag{27}$$

In [9], χ_{dk} corresponds to the expression $\mu(k) \cdot \sqrt{2\pi}$..

Altenbernd points out that approaches using e-functions yield similarly good results, since they are based on the Poisson distribution. Such approaches using e-functions are described by Macfadyen in [10], Subrahmanyam in [11] and Früchtenicht in [12]:

$$J(H) = \mu_0 \cdot \sum_{k=1}^{N} \chi_{dk} \cdot H_{\sigma k} \cdot \left(1 - e^{-\frac{H}{H_{\sigma k}}}\right) \tag{28}$$

$$\lim_{H \to \infty} J(H) = \mu_0 \cdot \sum_{k=1}^{N} \chi_{dk} \cdot H_{\sigma k} = J_{sat} \tag{29}$$

$$\chi_d = \frac{1}{\mu_0} \cdot \frac{dJ}{dH} = \sum_{k=1}^{N} \chi_{dk} \cdot e^{-\frac{H}{H_{\sigma k}}} \tag{30}$$

$$\chi_d(H = 0) = \sum_{k=1}^{N} \chi_{dk} = \mu_{ri} - 1 \tag{31}$$

By rearranging the terms, we obtain the form presented by El-Sherbiny in [13] and Hwang in [14]:

$$J = J_{sat} - \mu_0 \cdot \sum_{k=1}^{N} \chi_{dk} \cdot H_{\sigma k} \cdot e^{-\frac{H}{H_{\sigma k}}} \tag{32}$$

As an extension of the approximation for small field strengths, Macfadyen proposes an additional term:

$$J = \mu_0 \cdot \chi_{d0} \cdot H \cdot e^{-\frac{H}{H_{\sigma 0}}} + J_{sat} - \mu_0 \cdot \sum_{k=1}^{N} \chi_{dk} \cdot H_{\sigma k} \cdot e^{-\frac{H}{H_{\sigma k}}} \tag{33}$$

7 Approximations Using Splines and Appropriate Extrapolation

Splines are modeled after the long, elastic rulers used in shipbuilding.

Cubic splines interpolate $N+1$ knots $(x_k, y_k)|_{k=1,N+1}$ using N cubic polynomials:

$$s_k = c_{3k} \cdot (x - x_k)^3 + c_{2k} \cdot (x - x_k)^2 + c_{1k} \cdot (x - x_k) + c_{0k} \tag{34}$$

The derivatives are easy to find:

$$s_k{}' = 3 \cdot c_{3k} \cdot (x - x_k)^2 + 2 \cdot c_{2k} \cdot (x - x_k) + c_{1k} \tag{35}$$

$$s_k{}'' = 6 \cdot c_{3k} \cdot (x - x_k) + 2 \cdot c_{2k} \tag{36}$$

There are $4 \cdot N$ parameters tob e dtermined:

$$y_k = s_k(x_k) = c_{0k} \tag{37}$$

$$\begin{aligned} &y_{k+1} = s_k(x_{k+1}) = \\ &c_{3k} \cdot (x_{k+1} - x_k)^3 + c_{2k} \cdot (x_{k+1} - x_k)^2 + c_{1k} \cdot (x_{k+1} - x_k) + c_{0k} \end{aligned} \tag{38}$$

Slope and curvature should match at the interval boundaries:

$$\begin{aligned} &s_k{}'(x_{k+1}) = s_{k+1}{}'(x_{k+1}) \quad \rightarrow \\ &3 \cdot c_{3k} \cdot (x_{k+1} - x_k)^2 + 2 \cdot c_{2k} \cdot (x_{k+1} - x_k) + c_{1k} = c_{1k+1} \end{aligned} \tag{39}$$

$$s_k{}''(x_{k+1}) = s_{k+1}{}''(x_{k+1}) \rightarrow 6 \cdot c_{3k} \cdot (x_{k+1} - x_k) + 2 \cdot c_{2k} = 2 \cdot c_{2k+1} \tag{40}$$

There are various approaches to express the two missing equations of state:

- Natural boundary conditions: No curvature at either end

$$s_1{}^{''(x_1)} = 0 \tag{41}$$

$$s_N{}''(x_{N+1}) = 0 \tag{42}$$

- Clamped: The slopes on both ends $s_1{}'(x_1)$ and $s_N{}'(x_{N+1})$ are given.
- Periodic: Of course, $y_{N+1} = y_1$ must hold; in addition, the following must be satisfied:

$$s_1{}'(x_1) = s_N{}'(x_{N+1}) \tag{43}$$

$$s_1{}''(x_1) = s_N{}''(x_{N+1}) \tag{44}$$

For a description of additional boundary conditions and further details, please refer to the mathematical literature, including de Boor [15] and Dierckx [16].

7.1 Smoothing Splines

One drawback is the small but possible occurrence of unwanted roughness, especially when applied to raw data containing measurement or rounding errors. The desire for smoothing gives rise to the idea of approximation using smoothing splines: The distance to the control points is minimized, and roughness - as measured by the second derivative - is also minimized:

$$p \cdot \sum_{k=1}^{N+1} \left(y_k - f(x_k)\right)^2 + (1-p) \cdot \int_{x_1}^{x_{N+1}} f''(x) \cdot dx \tag{45}$$

The first term can be regarded as a measure of accuracy, and the second term as a measure of roughness. p is a suitable smoothing factor.

When applying this to magnetization curves, the coefficients c_{0k} must be corrected by subtracting c_{01} to ensure that - even though the origin $J_1(H_1 = 0) = 0$ is explicitly included in the control points - the approximation passes through the origin.

For the numerical determination of the coefficients, routines are available in, among others,

- Matlab https://de.mathworks.com/help/curvefit/csaps.html
- Octave https://octave.sourceforge.io/splines/function/csaps.html
- Python https://octave.sourceforge.io/splines/function/csaps.html

In this process, splines with natural boundary conditions (no curvature at either end) are used.

Ungethüm and Hülsebusch describe a Modelica implementation [18] in [17].

However, the extrapolation to high field strengths remains inadequate. It is unlikely that the magnetic field strengths in the measurement series will reach levels high enough for the polarization to approach the saturation polarization sufficiently. For some applications, therefore, an alternative extrapolation is essential, as described by Rao and Kuptsov in [19]:

7.2 Extrapolation to High Field Strength

An analysis of 10 sets of measurements (VoestAlpine: M270-50A, M310-50A, M330-50A, M350-50A, M400-50A, M470-50A, M530-50A, M600-50A, M700-50A, M800-50A) has shown that, starting at a suitably chosen field strength H_0, the measured values can be approximated with satisfactory accuracy by exponential extrapolation:

$$J(H) = J_0 + (J_{sat} - J_0) \cdot \left(1 - e^{-\frac{H-H_0}{H_{par}}}\right) \tag{46}$$

By minimizing the distances to the support points, one obtains the optimal parameter H_{par} as well as the saturation polarization J_{sat} - which is not known from the measurement data - with good accuracy.

The direct evaluation of the relative permeability is numerically unproblematic, since the region of low field strengths is excluded:

$$\mu_r(H) = 1 + \frac{J}{\mu_0 \cdot H} \tag{47}$$

$$\mu_{rd}(H) = 1 + \frac{1}{\mu_0} \cdot \frac{dJ}{dH} = 1 + \frac{J_{sat} - J_{k_B}}{\mu_0 \cdot H_{par}} \cdot e^{-\frac{H-H_0}{H_{par}}} \tag{48}$$

To achieve a smooth transition from the smoothing splines region J_{SS} to the exponential extrapolation region J_{EE}, a linear transition function h is defined for the interval $[H_1, H_2]$ that has to be selected:

$$h(H) = \max\left(0, min\left(1, \frac{H - H_1}{H_2 - H_1}\right)\right) \tag{49}$$

$$J(H) = (1 - h) \cdot J_{SS} + h \cdot J_{EE} \tag{50}$$

Saturation polarization is reached asymptotically. The field strength at which the deviation from saturation polarization has shrunk to a given accuracy k can be determined as follows:

$$\begin{aligned} &J(H_{sat}) = (1 - k) \cdot J_{sat} = J_0 + (J_{sat} - J_0) \cdot \left(1 - e^{-\frac{H_{sat} - H_0}{H_{par}}}\right) \rightarrow \\ &H_{sat} = H_0 + H_{par} \cdot ln\left(\frac{J_{sat} - J_0}{k \cdot J_{sat}}\right) \end{aligned} \tag{51}$$

8 Code and Example

Code running in Matlab and Octave calculating the parameters from raw data is provided in https://github.com/AHaumer/AppMag as well as an example of using the parameters in Modelica.

The code has been used on quality M330-50A (VoestAlpine) using the following parameters:

- $p = 0.005$
- $H_0 = 2\ \frac{kA}{m}$
- $H_1 = 5\ \frac{kA}{m}$
- $H_2 = 20\ \frac{kA}{m}$

The following results have been obtained:

- $\mu_{ri} = 1482.8$
- $\mu_{rMax} = 7256.7$
- $H_{\mu Max} = 97.0\ \frac{A}{m}$
- $B_{\mu Max} = 0.8847\ T$
- $J_{sat} = 1.9953\ T$
- $H_{sat} = 145.8\ \frac{kA}{m}$ to reach J_{sat} with $1\ ppm$ accuracy

A comparison between the approximated and the measured values is shown in Figure 2 and Figure 3.

The results of a simulation in Modelica using the previously obtained parameters is shown in Figure 4.

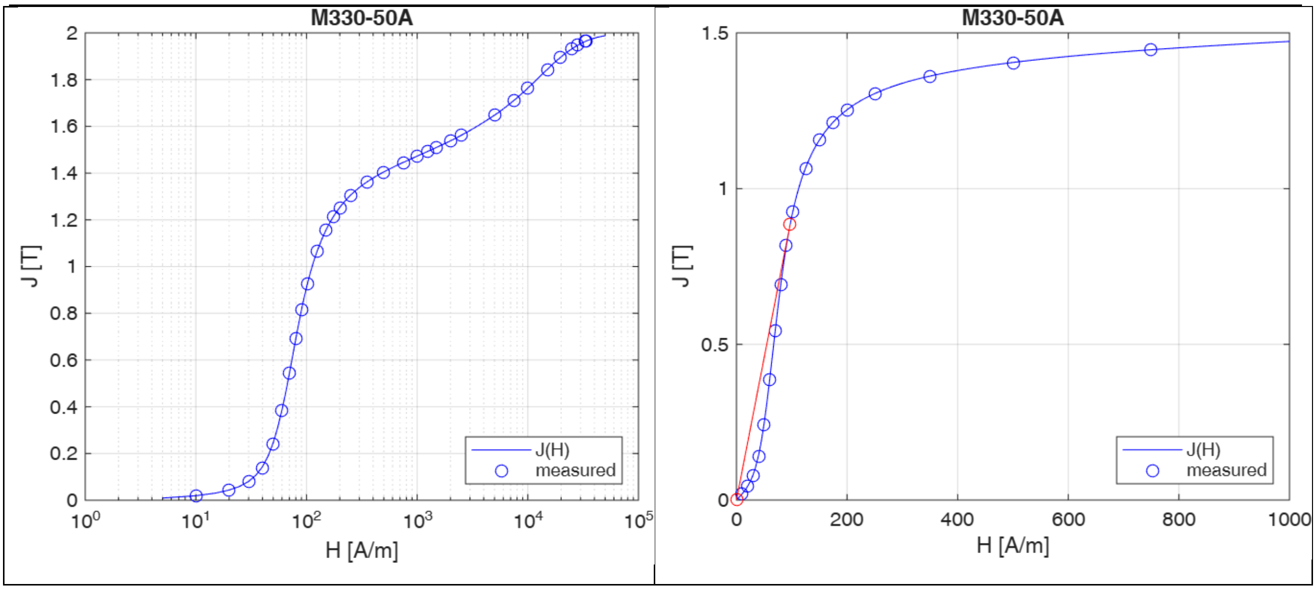

Figure 2 Polarization *J(H)* for the whole measured range and for the reagion of low field strength

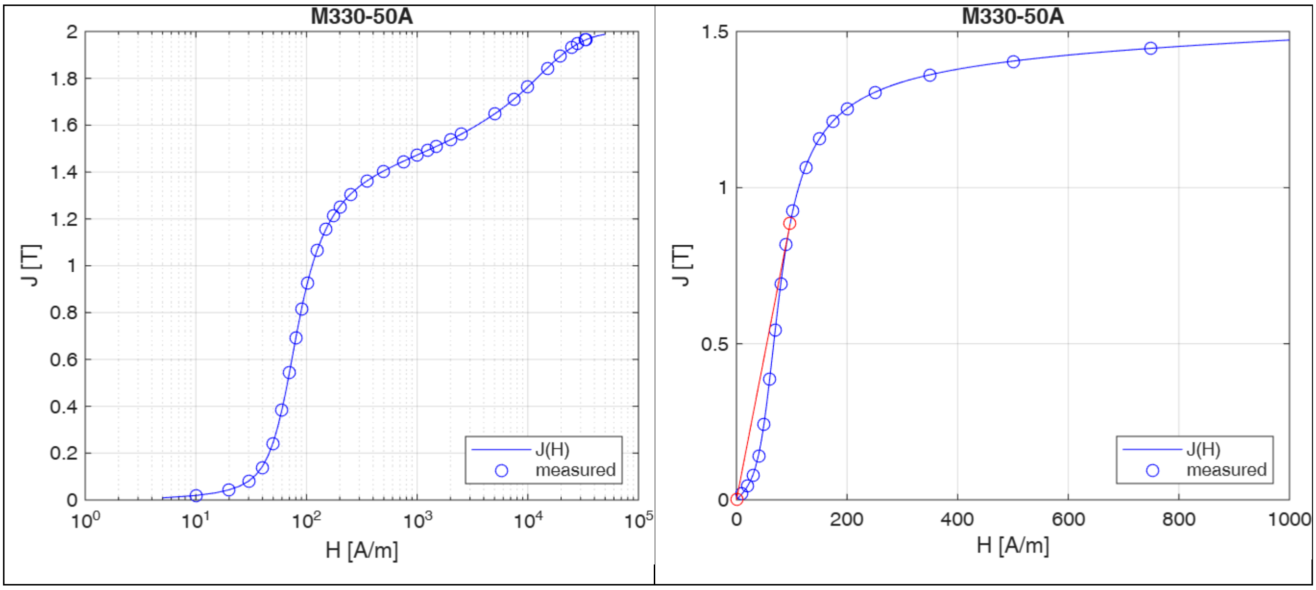

Figure 3 Permeability $\mu_r(H)$ and $\mu_{rd}(H)$ for the whole measured range and for the reagion of low field strength

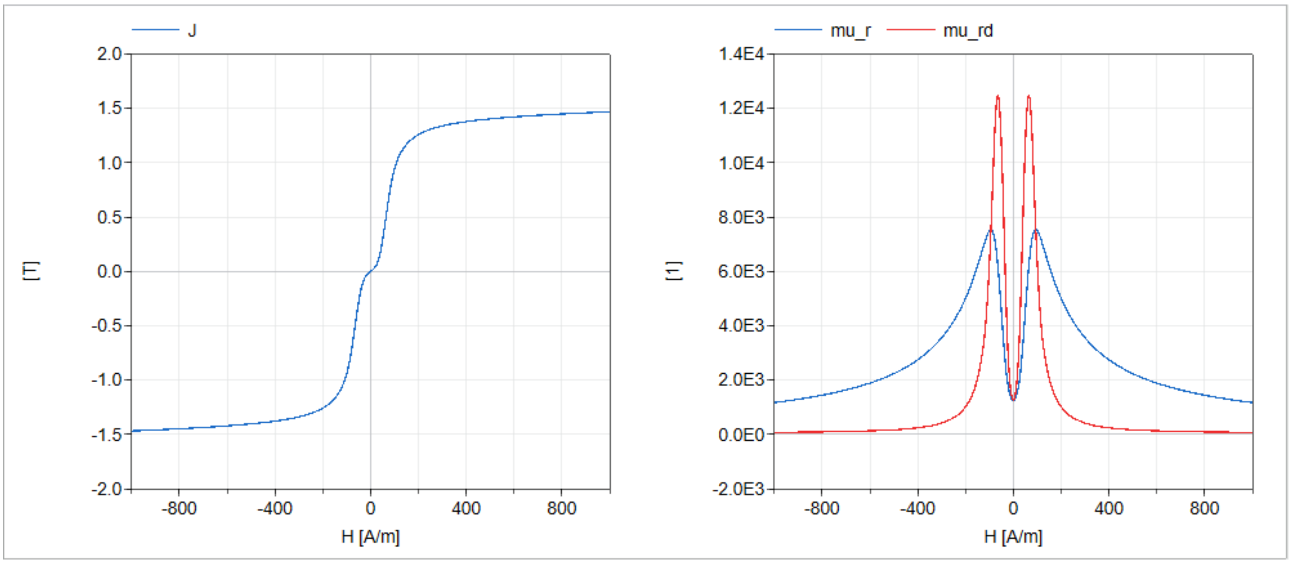

Figure 4 Simulation of M330-50A in Modelica

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